



Jian-Pi-Yi-Shen formula restores iron metabolism from dysregulation in anemic rats with adenine-induced nephropathy

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ABSTRACT

Ethnopharmacological relevance: Jian-Pi-Yi-Shen (JPYS) is a herbal decoction being used to relieve the symptoms of chronic kidney disease (CKD) and its complications, including anemia, for over twenty years. Nonetheless, it is unclear how JPYS influences renal anemia and iron metabolism.

Aim of the study: An analysis of network pharmacology, chemical profiling, and *in vivo* experiments was conducted to identify the impact of JPYS on JAK2-STAT3 pathway and iron utilization in renal anemia and CKD. **Materials and methods:** The chemical properties of JPYS and its exposed ingredients were detected *in vivo*. And based on the aforesaid chemical compounds, the potential targets and signaling pathways of JPYS for renal anemia treatment were predicted by network pharmacology. Afterward, an adenine-feeding animal model of CKD-related anemia was developed to verify the mechanism by which JPYS modulates iron recycling to treat renal anemia. Renal injury was estimated by serum creatinine (Scr), blood urea nitrogen (BUN), histopathological examinations and fibrosis degree. Western blot, enzyme-linked immunosorbent assay (ELISA), quantitative real-time polymerase chain reaction (qPCR) and immunohistochemistry approaches were utilized to assess the levels of JAK2, STAT3 and iron metabolism-related factors.

Results: There were 164 active ingredients identified in JPYS, including prototypes and metabolites *in vivo* were identified in JPYS, and 21 core targets were found through network pharmacology based on topological characteristics. Combined with the core targets and pathway enrichment analysis, the majority of the candidate targets were associated with the JAK2-STAT3 signaling pathways. Experimental results indicated that JPYS treatment significantly decreased the expression of BUN and Scr, restored renal pathological damage, down-regulated fibrosis degree, and improved hematological parameters such as red blood cell, hemoglobin and hematocrit in CKD rats. Furthermore, JPYS significantly restored iron metabolism from dysregulation by increasing the levels of iron and ferritin in the serum, inhibiting the production of hepcidin in liver and serum, and regulating transferrin receptor 1 in bone marrow. Meanwhile, the expression of JAK2 and STAT3 was suppressed by JPYS treatment.

Conclusions: Based on these results, JPYS reduces hepcidin levels by inhibiting the activation of JAK2-STAT3 signaling, thereby protecting against iron deficiency anemia.

Abbreviations: BP, Biological processes; BUN, Blood urea nitrogen; CC, Cellular components; CKD, Chronic kidney disease; DAB, Diaminobenzidine; DMT1, Divalent metal transporter 1; EPO, Erythropoietin; ESA, Erythropoiesis stimulating agents; GFR, Glomerular filtration rate; GO, Gene ontology; HCT, Hematocrit; HGB, Hemoglobin; HIF, Hypoxia-inducible factor; IOD, Integrated optical density; JPYS, Jian-Pi-Yi-Shen; PFA, Paraformaldehyde; RBC, Red blood cell; Scr, Serum creatinine; TCM, Traditional Chinese medicine; TfR1, Transferrin receptor 1.

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1. Introduction

Globally, 15–20% of adults suffer from chronic kidney disease (CKD), which became an important contributor to death rates due to its low glomerular filtration rate (GFR) and increased albuminuria. A 2020 assessment conducted by the Global Burden of Disease group found CKD as most common global factors that contribute to a bad prognosis (Chen et al., 2019; Deng et al., 2021; Matsushita et al., 2022). A major complication of CKD is anemia, which is characterised by low hemoglobin (HGB) and hematocrit (HCT) levels, as well as low red blood cell counts (RBC) (Koury and Haase, 2015). Renal anemia can hasten the progression of CKD while also causing neurocognitive impairment, sleep difficulties, cardiovascular comorbidities, and higher mortality (Li et al., 2016; Wu et al., 2022). The pathogenesis of CKD-associated anemia is multifaceted, with inadequate erythropoietin (EPO) expression and iron insufficiency being the two primary pathogenetic causes (Malyszko et al., 2019). A standard treatment for anemia in individuals with CKD is to administer erythropoiesis stimulating agents (ESA), which often required iron supplementation (Yap et al., 2021). However, the afore-said therapies may show signal for harm, such as increasing blood pressure, promoting tumor growth and increasing the risk of thromboembolism in patients with malignant tumors. The oral prolyl hydroxylase inhibitor of hypoxia-inducible factor (HIF) such as roxadustat is a newer class of agents that have been already approved for use in some countries to treat anemia, but long-term use may increase their cardiovascular risk in CKD patients (Barratt et al., 2021; Dhillon, 2019). Given the concern about safety with these treatments, the development of additional therapeutics alternative in the management of renal anemia are therefore still needed.

The practice of traditional Chinese medicine (TCM) has gained a great deal of experience in treating CKD as well as its associated diseases. Among many of them, Jian-Pi-Yi-Shen (JPYS) contains eight herbs and has been utilized for decades with good efficacy in clinical treatment of CKD and anemia (Fu et al., 2017; Liu et al., 2022; Yi et al., 2004). In TCM theory, renal anemia is considered as chronic kidney damage that leads to insufficiency of kidney essence and qi, and difficulty of nourishing the spleen and stomach, resulting in the deficiency of spleen and kidney. Thus anemia is caused by lack of source for blood and qi formation. The Compendium of Materia Medica has a high evaluation of Astragali Radix, which is considered as “the elder of tonic medicine”. JPYS uses Astragali Radix as the sovereign drug and possesses ability to make the kidney essence sufficient by strengthening the movement and transformation of spleen qi, thereby providing qi and blood with active biochemical sources. Collectively, JPYS can tonify kidney and strengthen essence, fortify the spleen and boost qi, replenish qi and nourish blood, support healthy energy, and finally relieve renal anemia and CKD (Li et al., 2022). Our earlier studies have demonstrated that JPYS could correct renal anemia primarily by enhancing EPO expression via HIF signaling (Wang et al., 2020). We also found that the dysfunction of iron recycling in CKD rats and the potential beneficial impact of JPYS on iron metabolism. However, the detail regulating mechanism of JPYS in the utilization of available iron stores in renal anemia and CKD should be explained in greater depth.

TCM has formed a comprehensive and unique medical system after being developed over thousands of years. However, lack of scientific research approach to explain the molecular pathways of TCM formula owing to its complicated chemical compounds within herbs. Hopkins (2007) proposed network pharmacology to investigate the multilayer interconnections of diseases, genes, and medications as a whole, which is important for identifying therapeutic targets and potential signaling pathways present in TCM.

Here, to further reveal the mechanism of JPYS in treating renal anemia, we first analyzed the chemical profile of JPYS and its exposed components *in vivo*. And then, JPYS components and their metabolites absorbed *in vivo* was performed by network pharmacology to predict the potential targets and pathways of JPYS in treating renal anemia. Finally,

CKD-associated anemia animal model was carried out to validate the mechanism of JPYS in treating renal anemia by regulating iron metabolism. These studies may provide a fresh insight into the mechanism of JPYS as well as evidence supporting for the clinical application of JPYS.

2. Materials and methods

2.1. JPYS preparation

Medicinal herbs for JPYS were gained from the Pharmaceutical Department of Shenzhen Traditional Chinese Medicine Hospital. Astragali Radix (Huang Qi, roots of *Astragalus membranaceus* var. *mongholicus* (Bunge) P.K.Hsiao) 30 g, Atractylodis Macrocephalae Rhizoma (Bai Zhu, rhizomes of *Atractylis macrocephala* (Koidz.) Hand.-Mazz.) 10 g, Dioscoreae Rhizoma (Shan Yao, rhizomes of *Dioscorea opposita* Thunb.) 30 g, Cistanches Herba (Rou Cong Rong, herbs of *Cistanche deserticola* Y.C. Ma) 10 g, Amomi Fructus Rotundus (Dou Kou, fruits of *Amomum kravanh* Pierre ex Gagnep.) 10 g, Salviae Miltiorrhizae Radix et Rhizoma (Dan Shen, roots and rhizomes of *Salvia miltiorrhiza* Bunge) 15 g, Rhei Radix et Rhizoma (Da Huang, roots and rhizomes of *Rheum palmatum* L.) 10 g, Glycyrrhizae Radix et Rhizoma (Gan Cao, roots and rhizomes of *Glycyrrhiza uralensis* Fisch.) 6 g were weighed and boiled twice for 1 h each time in 8 times of water. Our earlier research detailed the preparation and quality control of JPYS extract (Liu et al., 2018), and the extraction yield was ~31.85%. The spray-dried powder of JPYS was kept at −80 °C and re-dissolved with Milli-Q water by sonicating for 10 min, vortexing for 60 s and centrifugating to obtain the supernatant before the treatment.

2.2. Animals and treatment

Male Sprague-Dawley (SD) rats (6 weeks, 190–210 g) in good health were supplied by Guangdong Vital River (Foshan, China). During 1 week of acclimation, all animals were supplied with food and water without restriction. 24 rats were randomised into four groups: the control group (n = 6); the CKD group (n = 6); the low dose of JPYS treatment group (CKD + 0.98 g/kg JPYS, n = 6); and the high dose of JPYS treatment group (CKD + 3.92 g/kg JPYS, n = 6). 0.75% adenine diet induced CKD after 4 weeks (Liu et al., 2022). For four weeks, rats in the treatment groups received various doses of JPYS daily through gavage beginning in the third week of the adenine-rich diet. In both the control and CKD groups, rats were given double-distilled water *via* gavage. The control group showed normal changes in body weight. In contrast, the body weight of the CKD rats was robustly lower than the control group but up-regulated after receiving JPYS treatment.

2.3. Molecular network and chemical scaffold classification assay

Oral administration of JPYS extract was performed on rats, whereas the control group was given a solution of double-distilled water in the same amount. After the administration, urine and feces were collected from 3 rats in each group every 2 h by metabolic cages, and blood from ophthalmic veins was collected in tubes at intervals of 5, 15, 30, 60, 120, 240, 360, 480, and 600 min. And the remaining rats were euthanized through cervical detachment, and the bile was collected at 1, 3, 5, 7, 9, and 10 h after administration. Then, the JPYS extracts, mixed plasma, urine, feces and tissue were identified by UHPLC-QTOF/MS. Next, the integrated molecular network was obtained effectively by MetabolitePilot software 2.0 (www.metabolitepilot-software.com) (Wang et al., 2022; Zhou et al., 2020a). Finally, Cytoscape software 3.7.2 (www.cytoscape.org) was employed to build the molecular network.

2.4. Network pharmacology

Target prediction of the active components detected by UPLC-QTOF/MS with TCMIP v2.0 (<http://cn/www.tcmip.com>) and Swiss target

Prediction (<http://www.swisstargetprediction.ch/>). OMIM (<https://www.omim.org/>) and GeneCards (<https://www.genecards.org/>) platforms were used to predict disease targets with the key words “Renal Anemia” and “CKD Anemia”.

Using Draw Venn Diagram (<http://bioinformatics.psb.ugent.be/webtools/Venn/>) to take the intersection of JPYS active components targets and renal anemia disease targets. The probable core targets were identified for further study by selecting the target nodes with degree, betweenness, and closeness, which were higher than their respective median values in the PPI network. In order to better understand the common targets between JPYS and Renal Anemia, we imported them into the DAVID Bioinformatics Resources (<http://david.abcc.ncifcrf.gov/>) for gene ontology (GO) and KEGG enrichment analysis.

2.5. Pathology

Kidneys from rats of all the groups were embedded in paraffin after being preserved in 4% paraformaldehyde (PFA) for 24 h at 25 °C. Kidney sections were stained with PAS and Masson staining after being deparaffinized and rehydrated. Then, representative images were acquired by Axio Scope A1 (Zeiss, Oberkochen, Germany) and ZEN 2.3 software, and quantified by using Image J (NIH, Bethesda, MD). The criteria for scoring were as follows: 0 = no tubular injury; 1 = < 30%; 2 = 40%–60%; 3 = > 60% tubular injury.

2.6. Biochemical parameters examination

RBC, HCT, and HGB were analyzed by the automatic blood cell analyzer (Bowlinman Sunshine Technology, Beijing, China). Serum creatinine (Scr) and blood urea nitrogen (BUN) levels were detected by corresponding detection kit (StressMarq Biosciences, British Columbia, Canada). Serum Iron was determined by Micro Serum Iron Concentration Assay Kit (Solarbio, Beijing, China). Serum hepcidin, ferritin in serum, liver and spleen, and ferroportin in liver and spleen were determined by ELISA kits (USCN Life Science, Wuhan, China), respectively.

2.7. Quantitative real-time PCR

Total RNA was obtained from the Trizol reagent (Life technologies, USA) for the analysis of gene expressions in the livers and BM-MNCs. The concentration of RNA was determined using an Ultramicro spectrophotometer (Wuyi Technology, Nanjing, China). The primers were: 5'-CCCATCTATGAGGGTTACGC-3' (sense primer, S) and 5'-TTTAATGTCACGCACGATTTC-3' (anti-sense primer, AS) for rat β -actin (150 bp; NM_031144.3); 5'-AATGGTTCGTACAGCAGCAG-3' (sense primer, S) and 5'-CACGGAAGTAGTCTCCACGA-3' (anti-sense primer, AS) for rat Tfr1 (200 bp; NM_022712.1); 5'-GAAGGAAGCGAGACACCAAC-3' (sense primer, S) and 5'-GAGGCAGTGTGTTGAGAGGT-3' (anti-sense primer, AS) for rat hepcidin (130 bp; NM_053469.2); 5'-TTCCAGACGATGGTGCTTCT-3' (sense primer, S) and 5'-GTGGAATGTGGGAGGGAAGA-3' (anti-sense primer, AS) for rat DMT1 (94 bp; NM_013173.2). β -actin ($2^{-\Delta\Delta C_t}$) was utilized as the reference gene for determining relative gene expressions. Results are based on at least three independent experiments.

2.8. Western blotting

Proteins were extracted from bone marrow mononuclear cells, kidney cortexes and liver tissues. BCA (Biosharp, China) was employed to determine the protein concentration. Proteins were loaded and separated on 10% SDS-PAGE gels and incubated with primary antibodies for (STAT3, #9139S, CST; p-STAT3, #ab32143, Abcam; JAK2, #ab108596, Abcam; p-JAK2, #3776S, CST; β -actin, #TA-09, Zs-BIO) for a whole night at 4 °C. HRP-labeled secondary antibodies (1:10000) were

subsequently used to incubate the membranes at 37 °C for 2 h. Immuno-reactive bands were visualized with the help of a Hypersensitive ECL chemiluminescence Kit (Thermo, USA) and the Fusion imaging system (P&Q Science&Technology, China).

2.9. Immunohistochemistry

The samples were first cleaned in physiological saline, then fixed in paraformaldehyde at a concentration of 4% paraformaldehyde (PFA). Following the steps of dewaxing and antigen restoration, the slides were then exposed to 3% hydrogen peroxide for a period of 10 min. The sections were further incubated with the p-STAT3 primary antibodies (#ab32143, Abcam) for a whole night at 4 °C, and the following day, they were developed using SignalStain diaminobenzidine (DAB) substrate. Slides were viewed with an Axio Scope A1 (Zeiss, Oberkochen, Germany). Three rats in each group and three fields ($\times 400$) of each rat were observed in a blinded manner and quantified by Image J.

2.10. Statistical analysis

The data were showed as means $\pm$ standard error of the mean (SEM). To compare distinct groups for analysis, a one-way analysis of variance (ANOVA) with Tukey's post hoc test was utilized. $P < 0.05$ was regarded as statistically significant.

3. Results

3.1. Molecular network profiling and chemical scaffold classification of absorbed compounds in JPYS

JPYS extract was first chemically analyzed by UPLC-TOF/MS (Fig. S1), which can ensure the quality of JPYS extract and further facilitate the characterization of JPYS compounds and their metabolites *in vivo*. To explore the *in vivo* exposure on absorbed components, MS information of different groups including normal and JPYS-treated animal were analyzed on the MetabolitePilot software 2.0. An integrated molecular network data of JPYS extracts, blank bio-samples such as urine, plasma, feces and bile, and JPYS-loaded bio-samples such as urine, plasma, feces and bile, was identified and constructed based on the UHPLC-QTOF/MS data. As shown in Fig. 1A, the molecular networking map showed the relationship between chemicalome, prototypes and metabolites of JPYS in rat bio-samples. The structural categorization of these components was shown in Fig. 1B. The findings revealed the potentially efficient compounds, including 45 anthraquinones, 50 flavonoids, 32 phenylpropanoids, 9 chromones, 3 lactones, 6 phenanthraquinones and 10 triterpenoid saponin. Scaffold cluster analysis indicates the structure classification of each component, as depicted in Fig. 1C. The prototypes of JPYS in rat bio-samples from different herbs was shown in Fig. 1D. Collectively, a total of 164 JPYS components including prototype and metabolite in rat bio-samples were identified (Table S1&2), which was conducted to further network pharmacology analysis.

3.2. Network pharmacology predicted the potential mechanism of JPYS in treating renal anemia

After UniProt standardization and deduplication, 180 drug-disease intersection gene targets were obtained for further analysis. We performed cluster analysis of the PPI network, screened out the top 21 targets. Accordingly, these 21 core targets, namely, STAT3, SRC, PIK3CA, HSP90AA1, TP53, MAPK1, AKT1, PTPN11, VEGFA, JUN, CREBBP, IL6, EGFR, ESR1, STAT1, JAK2, JAK1, NFKB1, IL2, ITGB3 and MAP2K1, ranked by degree value, were obtained, as shown in Fig. 2.

The functional enrichment analysis of the relevant targets genes from PPI network was employed to determine their potential biological functions in the therapy of renal anemia. Top 10 significant enriched GO

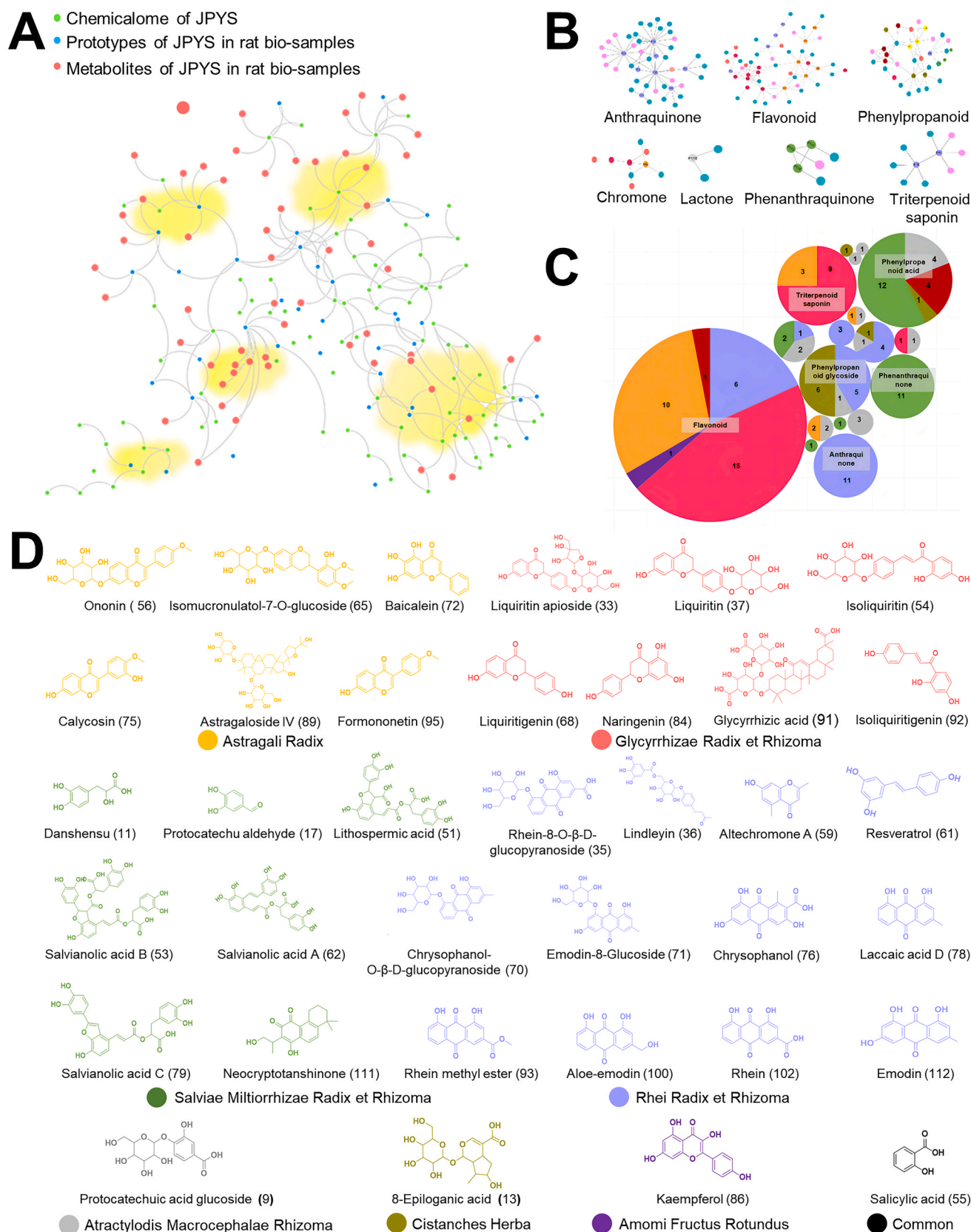


Fig. 1. Chemicalome-metabolome analysis of JPYS.

(A) Network of chemicalome-metabolome matching of JPYS. (B) Clustering network analysis for the components extracted in bio-samples. (C) Clustering circle map of components of JPYS. (D) The key chemical structures of potential active ingredients in JPYS.

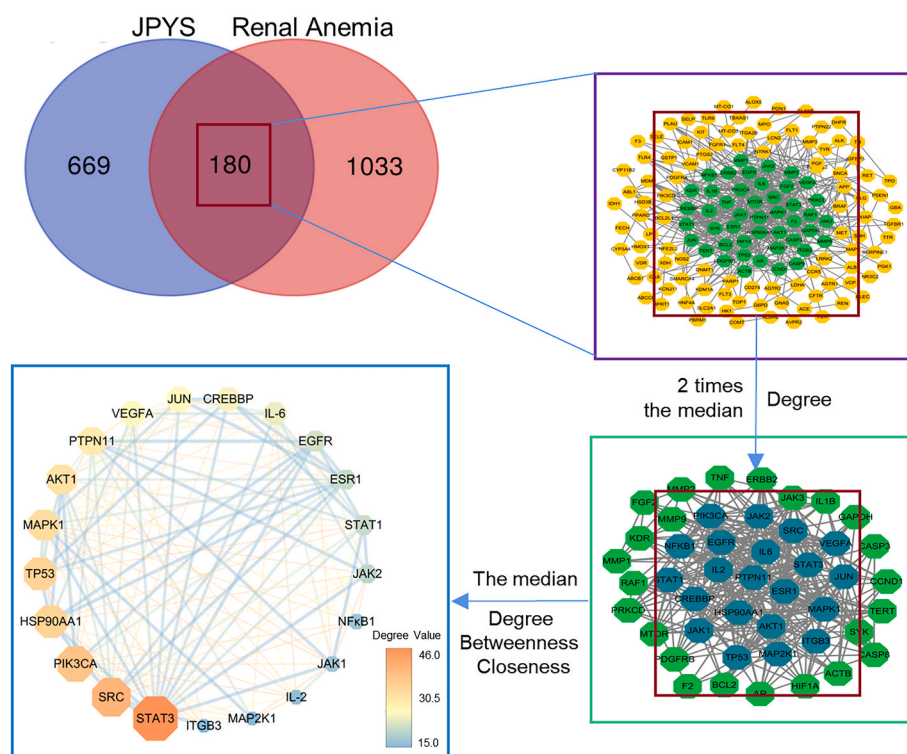


Fig. 2. The PPI network of JPYS and renal anemia targets.

After taking the intersection of the active ingredients of JPYS *in vivo* and renal anemia-related targets, the color of the targets changed from yellow to green according to the degree, betweenness and closeness values. The purple box showed all the targets, the green box showed the targets whose degree value was greater than 2 times the median, and the blue box showed the targets whose degree, betweenness and closeness values were greater than the median, which was the 21 core targets of this study. The color and size of the node correlated with its degree value. The thickness of the edge denoted the combined score. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

terms in BP, CC, and MF were shown (Fig. 3A). At the cellular level, for GO-BP category, the gene targets including cytokine-mediated pathway and positive regulation of gene expression were dramatically enriched. The plasma membrane and extracellular space were the two most GO-CC terms. These genes were markedly implicated in protein binding and identical protein binding for the GO-MF category.

To explore the function of JPYS in treating renal anemia, KEGG enrichment was conducted. Here, the top 25 signal pathways such as JAK-STAT and PI3K-AKT pathways were found, as indicated in Fig. 3B. The top 35 signaling pathways were involved in five main aspects, including regulation of iron metabolism, anti-inflammatory, regulation of cell proliferation, differentiation and apoptosis, anti-fibrotic and improving immune system function. Fig. 3C showed the “Drug-core components-key targets-pathways-pharmacological actions-diseases” interaction network, among which STAT3 and JAK2 were identified as targets with a relatively high degree of involvement in iron metabolism. We thus speculated that the therapeutic effects of JPYS on renal iron deficiency anemia might be involved in the JAK-STAT signaling pathway.

3.3. JPYS improved kidney function and pathological damage in CKD rats

The expression of BUN and Scr were significantly increased by more than two times in CKD rats compared to normal rats, indicating the renal function insufficiency. JPYS treatment significantly lowered elevated BUN and Scr levels in CKD rats in a concentration-dependent manner (Fig. 4A&B). Moreover, the renal pathological changes were investigated by PAS and Masson staining. CKD rats had the dilation of renal tubules and tubular cell necrosis according to PAS staining. Masson staining revealed that the fibrotic area was significantly enlarged in CKD rats as compared with normal rats. These pathological injuries improved after receiving JPYS treatment (Fig. 4C&D). These results indicate that JPYS has a renoprotective effect against CKD.

3.4. JPYS corrected anemia in CKD rats

The expression of RBC, HGB, and HCT in CKD rats was robustly lower than in normal rats, confirming the cause of anemia in CKD rats. JPYS at dose of 3.92 g/kg significantly increased the decreased levels of RBC, HGB and HCT, up to near normal levels; while JPYS at 0.98 g/kg showed an upward trend in aforesaid parameters, but not in a significant difference (Fig. 5A–C). It is believed that iron shortage contributes to anemia associated with CKD because it limits the amount of iron that is available for the process of erythropoiesis. Furthermore, to explore the iron status in CKD rats, the levels of serum iron and ferritin were detected. The results showed that rats with CKD revealed a substantial decrease in the amount of serum iron and serum ferritin as compared with normal rats, while JPYS treatment markedly increased the declined serum iron and serum ferritin levels (Fig. 5D&E). Besides, the level of serum hepcidin was also shown to be significantly increased in CKD rats, and JPYS significantly lowered serum hepcidin level in rats with CKD when compared to untreated rats, as shown in Fig. 5F. These data show that JPYS can alleviate the anemia and iron deficiency of CKD rats.

3.5. JPYS restored iron metabolism from dysregulation in CKD rats

Hepcidin is a major regulator of iron availability, and targeting hepcidin to restore iron metabolism from dysregulation is critical in the therapy of renal anemia. Here, we explored whether the JPYS-mediated hepcidin suppression was account for iron transport from liver and spleen to the plasma. In liver and spleen tissue, we found that the amount of ferritin was increased in CKD rats, but the amount of which was declined by treatment with JPYS (Fig. 6A). In parallel, the expression of ferroportin was strongly suppressed in CKD group but increased in JPYS group (Fig. 6B). Moreover, the mRNA expression of transferrin receptor 1 (TfR1) was increased in liver and spleen tissue in CKD group and was repressed in a concentration dependent manner in JPYS treatment group. In contrast, the mRNA expression of TfR1 in bone marrow-derived mononuclear cells showed the opposite trend (Fig. 6C). JPYS treatment inhibited the mRNA expression of hepcidin in liver (Fig. 6D),

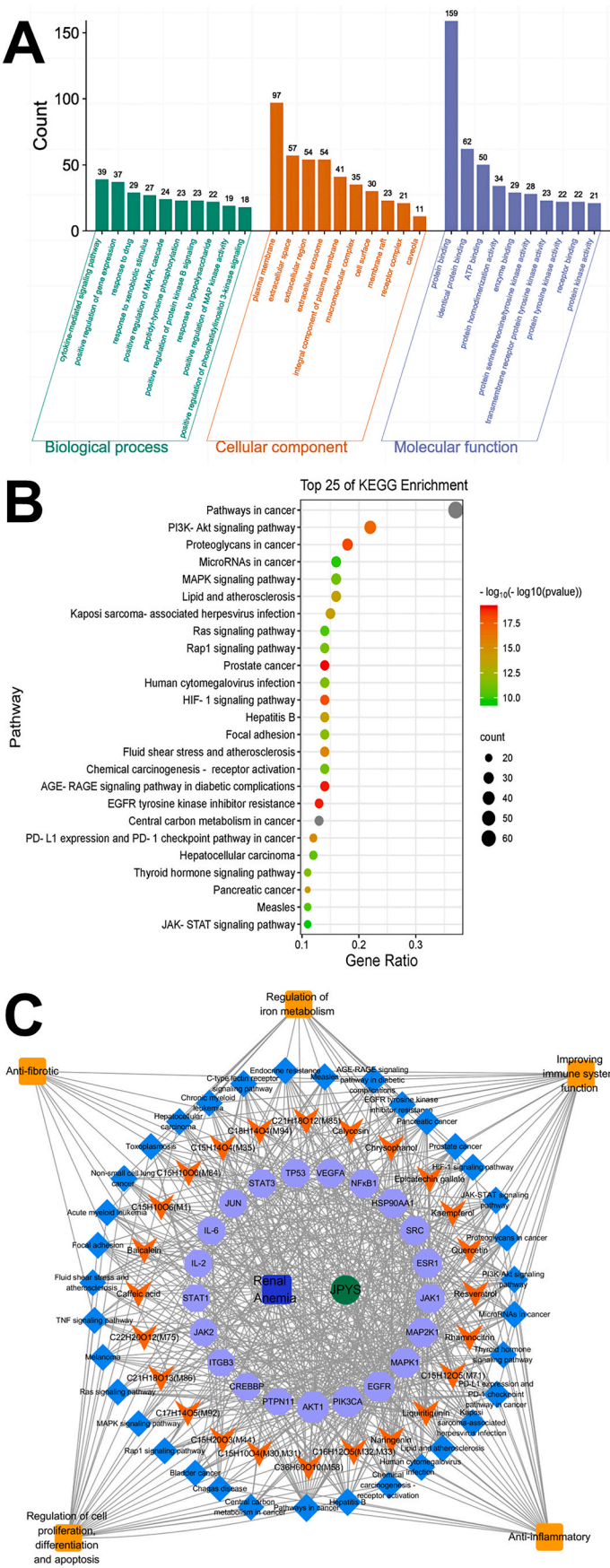


Fig. 3. Enrichment analysis and the underlying mechanisms of JPYS in treating Renal Anemia. (A) The top 10 significantly enriched GO-BP/MF/CC terms were displayed. Abbreviations: GO: Gene Ontology, BP: biological process, CC: cellular component, MF: molecular function. (B) The top 25 significantly enriched terms in KEGG pathways were displayed. (C) The underlying mechanisms of JPYS treatment of renal anemia via network analysis. Illustration of the relevance among active components of JPYS, 21 core targets, the top 35 of pathways and the corresponding pharmacological actions. Dark blue nodes referred to renal anemia, Green nodes referred to the JPYS, Purple nodes referred to the core targets, Orange nodes referred to the active ingredients of JPYS, Blue nodes referred to the signaling pathways, Yellow nodes referred to the pharmacological actions. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

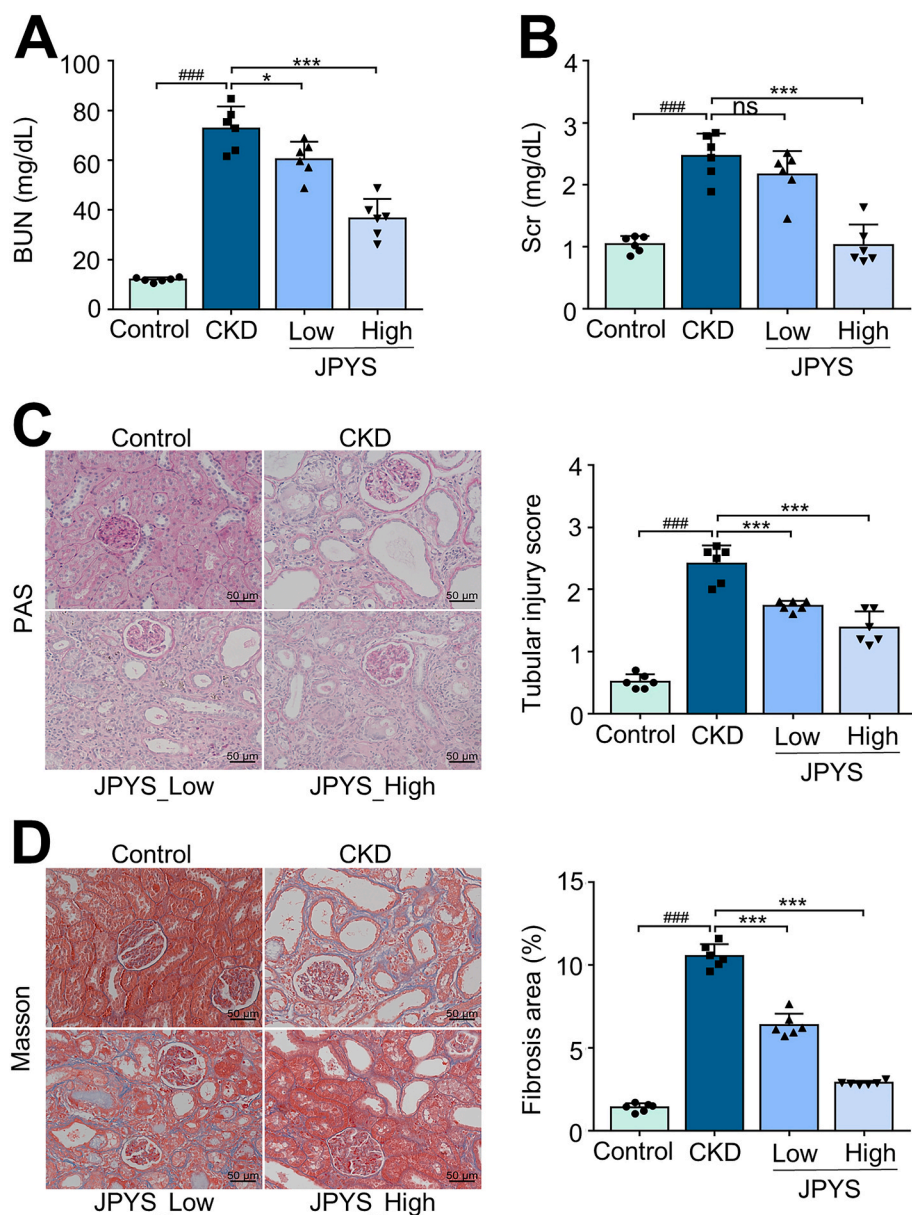


Fig. 4. Effects of JPYS on hematological parameters and kidney function in rats.

(A) Blood urea nitrogen levels. (B) Serum creatinine levels. Values are expressed as mean $\pm$ SEM, $n = 6$ rats per group (* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ between the indicated two groups). (C) Representative PAS staining images and quantitative analysis of tubular injury. (D) Representative Masson staining images and quantitative analysis of fibrosis. All images are shown at identical magnification, $\times 400$, scale bar = 20 μm . Values are expressed as mean $\pm$ SEM, $n = 6$ (### $p < 0.001$ compared with the control group; * $p < 0.05$ and *** $p < 0.001$ compared with the CKD group).

which was in line with the results shown in Fig. 5F. Besides, the mRNA expression of divalent metal transporter 1 (DMT1) was increased in spleen tissue in CKD group and was down regulated in JPYS treatment group (Fig. 6E).

3.6. JPYS induced protein phosphorylation of JAK2 and STAT3

To explain the potential mechanism of JPYS in attenuating hepcidin level, the protein expressions of p-JAK2 and p-STAT3 in the liver were detected by Western blot. As shown in Fig. 7A, the phosphorylation of JAK2 and STAT3 in CKD rats were induced more than twice that of the control group. After administration of JPYS, the phosphorylation levels of JAK2 and STAT3 were significantly decreased in a concentration dependent manner. We further detect the amount of p-STAT3 that translocated into the nucleus in liver cells by immunohistochemistry (Fig. 7B). We found that p-STAT3 level in the nucleus was higher in CKD rats compared with control group. After the treatment of JPYS concentration-dependently decreased the level of p-STAT3 in CKD rats. The findings suggest that JPYS may modulate iron metabolism via

inhibiting JAK2-STAT3 signaling, and which also verifies the prediction of network pharmacology analysis.

4. Discussion

A common cause of anemia among CKD patients is iron deficiency (Zhong et al., 2020), thus ensuring the iron availability for erythropoiesis can be therapies to correct iron deficiency anemia in CKD. In the current study, we studied the effects of JPYS in restoring iron metabolism from dysregulation, and the possible mechanism involving in JAK2-STAT3-mediated hepcidin level was also been studied in rat model of CKD induced by adenine (Fig. 8). The findings showed that JPYS could improve renal anemia by suppressing the activation of the JAK2-STAT3 signaling and modulating iron metabolism by reducing the level of hepcidin, which offers a new strategy for the research of the molecular function of JPYS in CKD renal anemia.

Anemia is frequently prevalent among CKD patients and mainly results from iron deficiency and erythropoiesis (Chifotides et al., 2022; Yun and Vincelette, 2015). CKD-associated anemia may exhibit

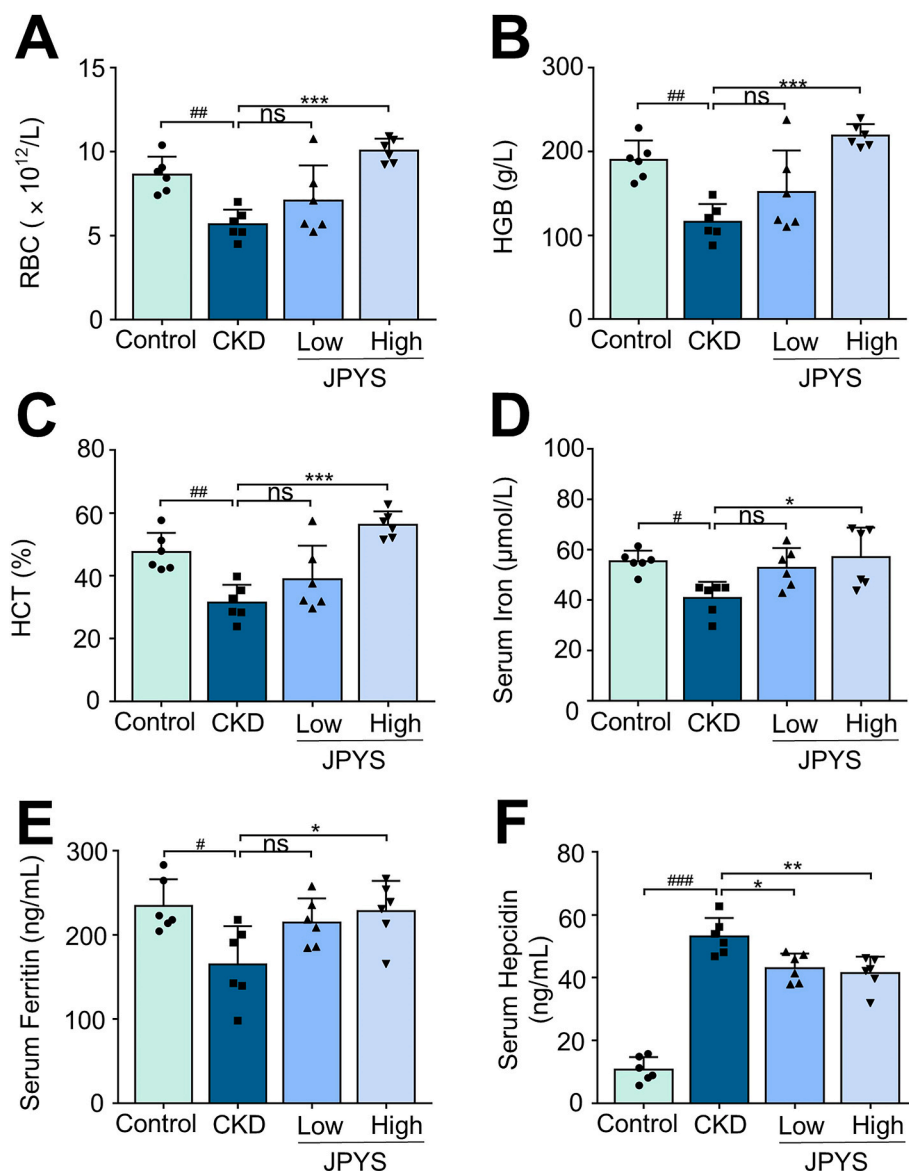


Fig. 5. Effects of JPYS on anemia and iron cycle in CKD rats.

(A) Red blood cell levels. (B) Hemoglobin levels. (C) Hematocrit levels. (D) Serum iron levels. (E) Serum ferritin levels. (F) Serum hepcidin levels. Values are expressed as mean $\pm$ SEM, $n = 6$ (# $p < 0.05$, ## $p < 0.01$ and ### $p < 0.001$ compared with the control group; * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ compared with the CKD group). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

functional iron deficiency. Studies have shown that inflammatory cytokines impaired erythropoiesis and restricted the supply of iron for erythropoiesis (Wang and Babitt, 2016). Our current studies showed that adenine feeding rats are related to anemia and iron shortage, as well as the level of serum hepcidin increased in CKD rats that might block iron availability. In support of this, some academics have proposed that hepcidin is the main regulator of iron uptake and release in tissues, ensuring erythron has a constant supply of iron. Hepcidin prevents iron transport by binding to ferroportin, a known iron exporter located on the basement membrane of hepatocytes, reticuloendothelial cells and enterocytes (Zhang et al., 2011). Ferroportin binds to hepcidin causing internalization from the plasma membrane into the cell and its eventual degradation and iron cannot be absorbed or recycled from the reticuloendothelial cells, resulting in reduced circulating iron levels, thereby inhibiting iron entry into the bloodstream (Gafer-Gvili et al., 2019). Novel therapies for alleviating iron deficient anemia in CKD patients that target hepcidin to increase available iron therefore are of great interest. In present studies, JPYS was found to decrease hepcidin level, promote the expression of ferroportin and improve the serum iron and serum ferritin of adenine-induced CKD rat. In addition, JPYS could decrease the expression of ferritin in tissues, promote the mobilization and utilization of iron, and thus improve renal anemia. Consistent with

this, our results have showed that the combination of Astragali Radix and Angelicae Sinensis Radix could prevent iron-deficiency anemia (Huang et al., 2016). Additionally, it should be noted that DMT1 is a significant indication of spleen iron recycling from RBC degradation (Berradi et al., 2008), and TfR1 is the major iron transporter (Valerio, 2007). Our study suggested that JPYS could decrease DMT1 synthesis in spleen and lowered TfR1 expression, reduce the storage of iron in the liver and spleen and promote the utilization of iron.

Network pharmacology is a multidisciplinary study field that combines computational and experimental methodologies to create new discoveries by integrating a vast quantity of information (Han and Han, 2021). The common method to get the active ingredients and the corresponding targets of the compound in the drug is directly from the database, however their accuracy and integrity cannot be guaranteed (Yuan et al., 2022; Zhou et al., 2020b). Therefore, we performed the *in vitro* and *in vivo* composition analysis of JPYS by UHPLC-QTOF/MS, and then detected the corresponding targets from the database, which made the results more reliable and persuasive. To accommodate the “multi-target, multi-channel, and multi-link” characteristics of Chinese herbal prescription, we shifted the study paradigm from the traditional “one target, one drug” mode to a new “network target, multi-components” approach.

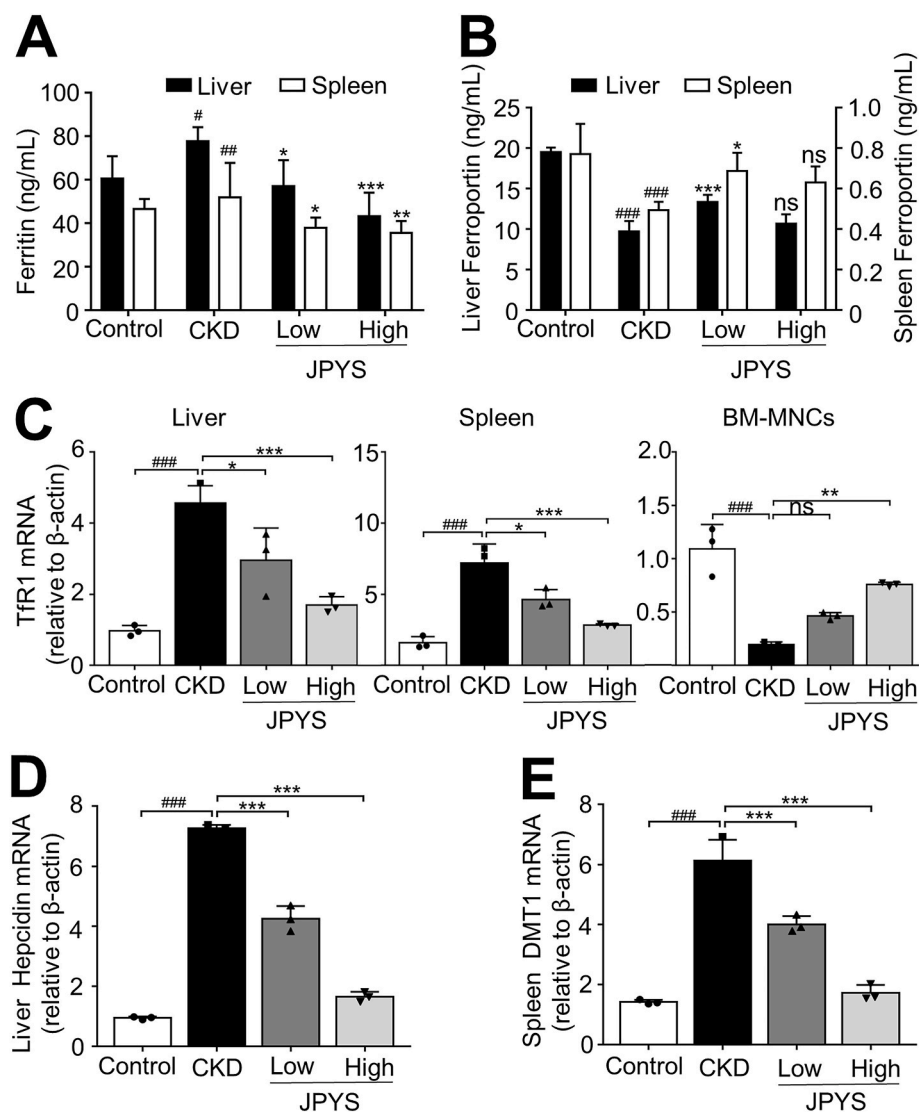


Fig. 6. Effects of JPYS on iron metabolism in CKD rats with renal anemia. **(A)** Ferritin expression in spleen and liver ($n = 6$). **(B)** Ferroportin expression in spleen and liver ($n = 6$). **(C)** The mRNA expression of TfR1 relative to β -actin by q-PCR were measured in the liver, spleen and BM-MNCs ($n = 3$, each with triplicate samples). **(D)** The mRNA expression of hepcidin relative to β -actin by q-PCR in liver ($n = 3$, each with triplicate samples). **(E)** The mRNA expression of DMT1 relative to β -actin by q-PCR in spleen ($n = 3$, each with triplicate samples). Values are expressed as mean $\pm$ SEM (# $p < 0.05$, ## $p < 0.01$ and ### $p < 0.001$ compared with the control group; * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ compared with the CKD group).

According to the report that STAT3 pathway is the key pathway in the restoration of iron metabolism dysregulation (Wang et al., 2018) and JAK2 is the key target in STAT3 pathway in renal anemia. After being activated, JAK2 phosphorylates the transcription factor STAT3, which then cause it to become active. Then, STAT3 moves into the nucleus of the cell, where it then binds to the canonical STAT3 binding site that is located in the proximal portion of the hepcidin promoter. This action causes an increase in the amount of hepcidin mRNA expression, which coincides with an increase in hepcidin protein production (Sun et al., 2012). In consistent with this, our study here showed that JPYS could decrease hepcidin expression by inhibiting JAK2-STAT3 signaling pathway. Furthermore, He et al. (2022) reported that *Salvia Miltiorrhiza* possessed anti-anemia effects by regulating JAK-STAT signaling. In addition, research has shown that cytokines, such as interleukin (IL)-6 and IL-1 β increase hepcidin expression through the JAK-STAT3 pathway, and inflammatory cytokines have a detrimental influence on iron homeostasis (Ueda and Takasawa, 2018; Weiss and Goodnough, 2005). Our previous studies also found that JPYS had bi-directional immune-modulatory effects, which can regulate pro-inflammatory cytokines level in RAW 264.7 cells under various conditions (Chen et al., 2018).

5. Conclusion

In conclusion, our findings showed that JPYS therapy improved renal anemia in CKD rats. Therapeutic effect of JPYS may be associated with the inhibition of JAK2-STAT3 signaling activation and the reduction of hepcidin to regulate iron metabolism. These findings can provide evidence for the therapeutic application of JPYS in the management of renal anemia in CKD.

Author contributions

Changhui Li: Data curation, Methodology, Formal analysis, Software, Writing-original draft, Writing – review and editing. Haipiao Huang: Methodology, Formal analysis, Data curation. Rui Wang: Methodology. Chi Zhang: Formal analysis. Shiyong Huang: Writing-original draft. Jinru Wu: Data curation. Pingli Mo and Huimin Yu: Software. Shunmin Li: Conceptualization, Writing-review and editing, Project administration. Jianping Chen: Project administration, Methodology, Writing-review and editing.

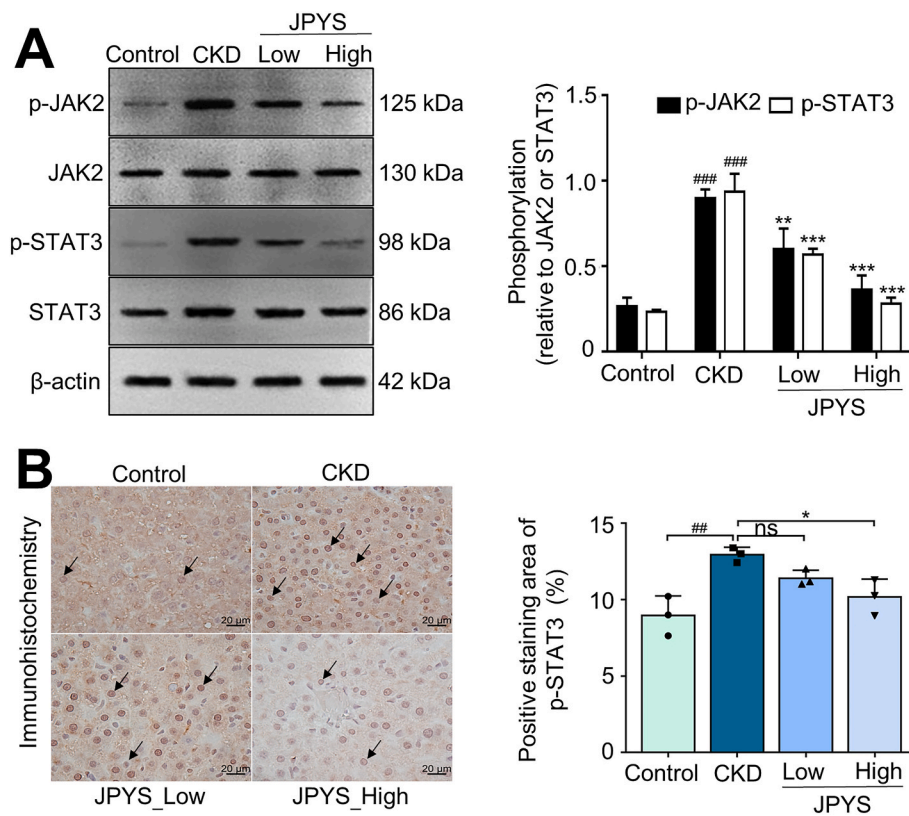


Fig. 7. Effects of JPYS on p-JAK2 and p-STAT3 protein expression in rats.

(A) Protein expression of p-JAK2, JAK2, p-STAT3 and STAT3 in liver. (B) Representative immunohistochemistry images of p-STAT3 expression in the nucleus in liver. All images are shown at identical magnification, $\times 400$, scale bar = 20 μ m. The arrows represented p-STAT3 in the nucleus. Quantitative analysis of positive staining areas for p-STAT3 ($n = 3$). Values are expressed as mean $\pm$ SEM (### $p < 0.01$ and ### $p < 0.001$ compared with the control group; * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ compared with the CKD group).

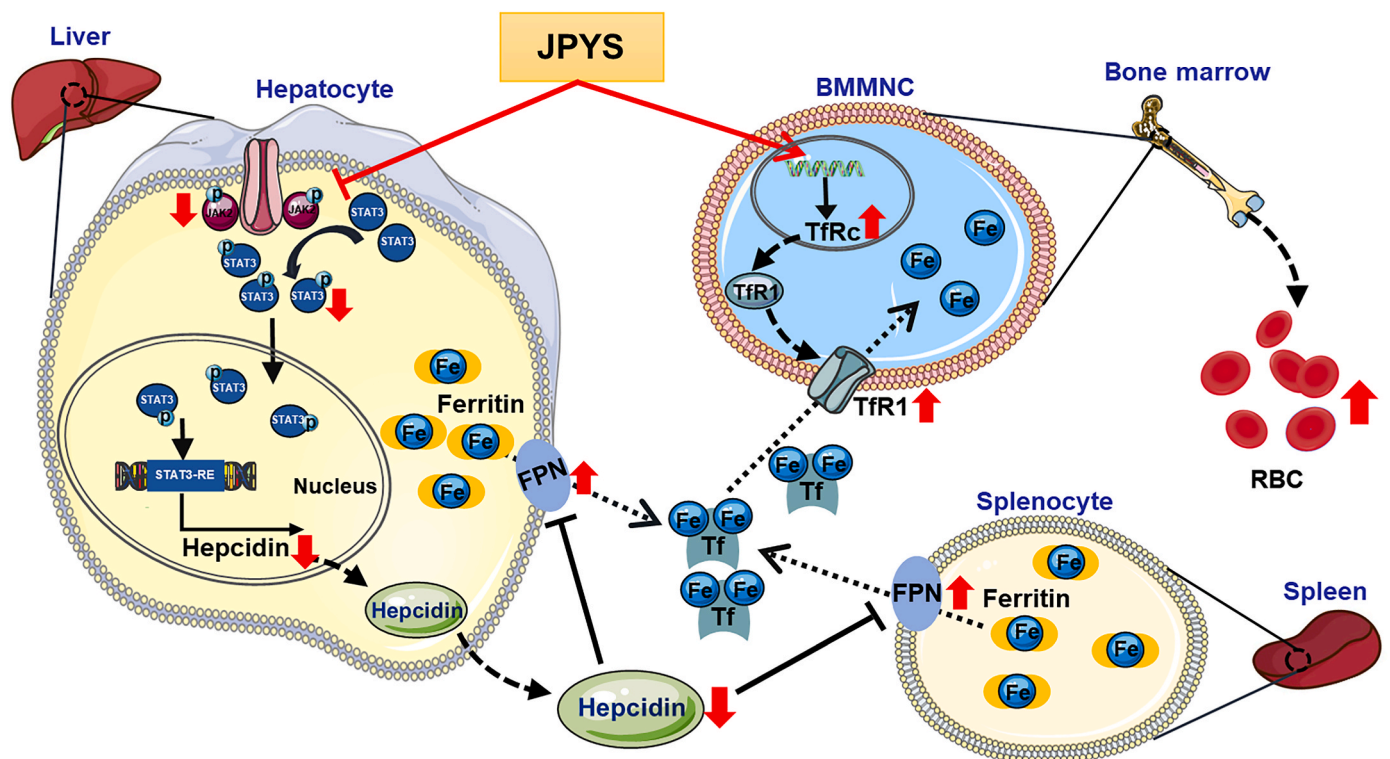


Fig. 8. Graphic representation of the underlying mechanism of JPYS in treating renal anemia.

JPYS could suppress the activation of JAK2-STAT3 signaling pathway, thereby inhibiting the expression of hepcidin, up-regulating FPN, and meanwhile increasing transferrin receptor 1, finally modulating iron metabolism and improving renal anemia.

Ethics approval and consent to participate

The study was approved by Ethics Committee for Animal Experimentation of China Technology Holdings (Shenzhen) Co.,Ltd. for the animal protocol (application No. 20220081).

Declaration of competing interest

The authors declare no conflict of interests.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.116526>.

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